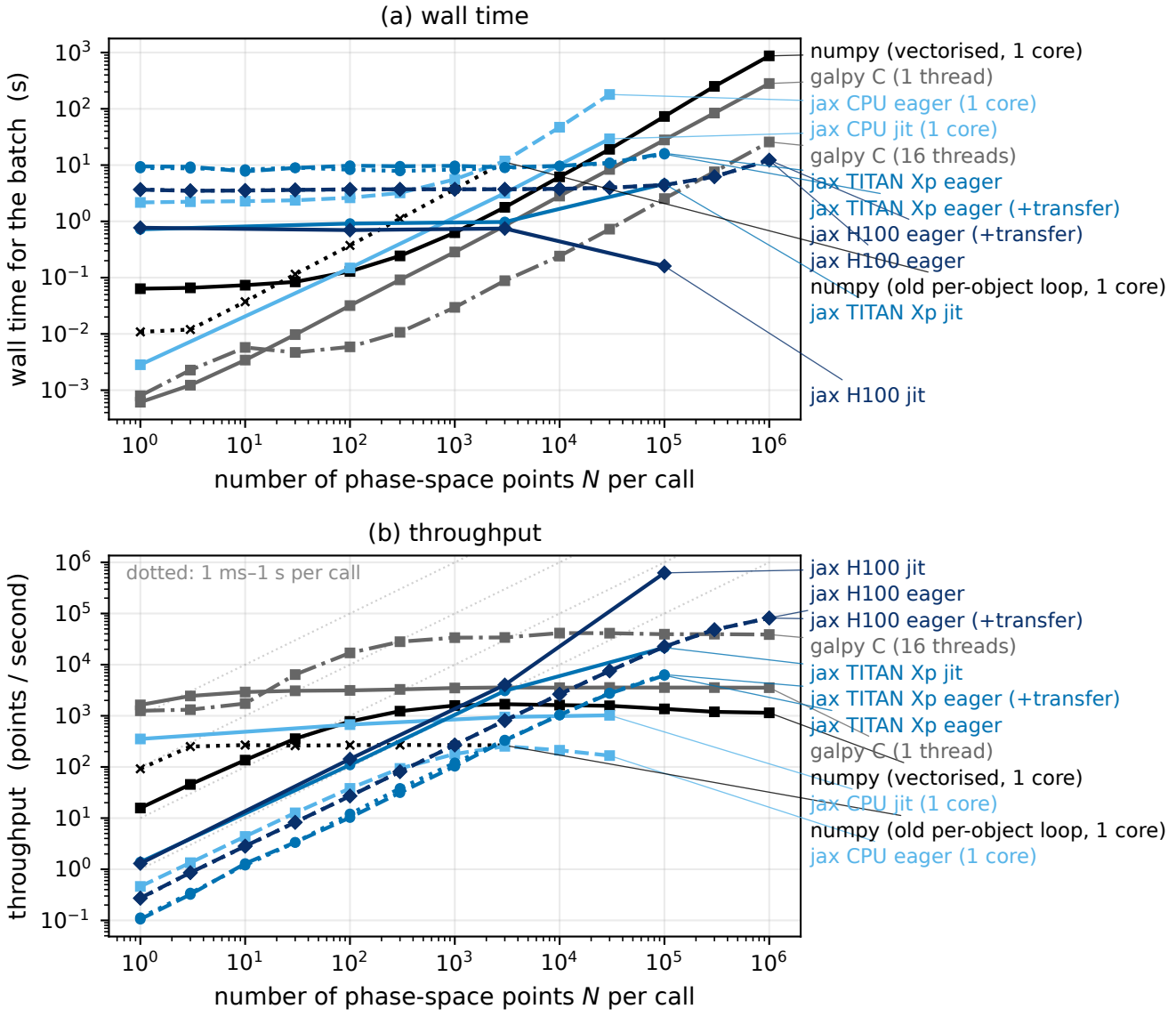
(a) wall time
wall time for the batch (s)
number of phase-space points N per call
numpy (vectorised, 1 core)
galpy C (1 thread)
jax CPU eager (1 core)
jax CPU jit (1 core)
galpy C (16 threads)
jax TITAN Xp eager
jax TITAN Xp eager (+transfer)
jax H100 eager (+transfer)
jax H100 eager
numpy (old per-object loop, 1 core)
jax TITAN Xp jit
jax H100 jit
(b) throughput
throughput (points / second)
dotted: 1 ms–1 s per call
number of phase-space points N per call
jax H100 jit
jax H100 eager
jax H100 eager (+transfer)
galpy C (16 threads)
jax TITAN Xp jit
jax TITAN Xp eager (+transfer)
jax TITAN Xp eager
galpy C (1 thread)
numpy (vectorised, 1 core)
jax CPU jit (1 core)
numpy (old per-object loop, 1 core)
jax CPU eager (1 core)